OntoLM Tutorial | Chapter 2 of 8

ISWC BARI 2026

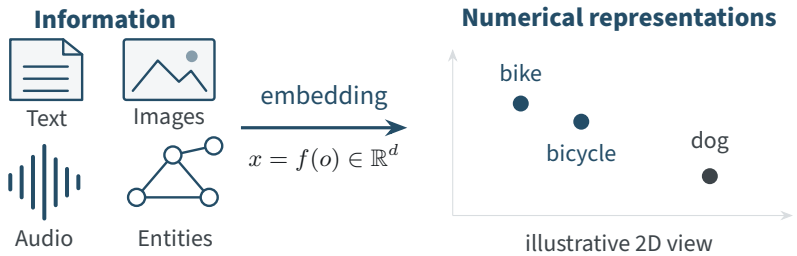
Geometric Embeddings: From Hierarchies to Axioms

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Embeddings map information into vectors



An embedding represents an object as a vector, preserving task-relevant information

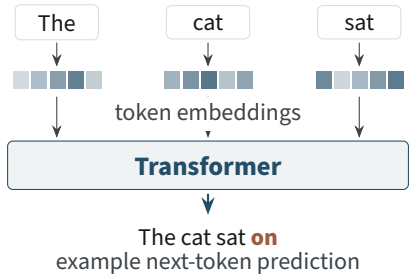
Embeddings support many different tasks

Simple: nearest-neighbour search



- Embed query and product descriptions
- Rank by vector similarity
- Retrieve by meaning, not exact wording

Complex: Transformer language models



- Map each token to a vector
- Use context to update representations
- Generate, translate and summarise

Roadmap: from hierarchies to ontologies

Three parts: hierarchies $\rightarrow$ ontologies $\rightarrow$ open problems

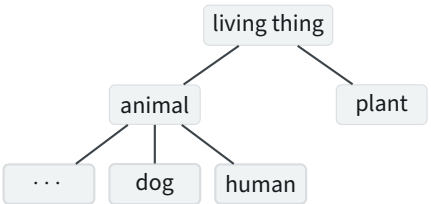
	Part	What it covers	Methods	Time
1	**Hierarchy embeddings**	is-a: asymmetry + transitivity, in low dimensions	order & box embeddings, hyperbolic space & cones, RegD[17]	20 min
2	**EL ontology embeddings**	beyond is-a: conjunction $\sqcap$, existential $\exists r$, plus analysis and training	ELEm[18], ELBE[20], BoxEL[21], Box^{2}EL[22], TransBox[23]	16 min
3	**Other works**	other logics, theoretical analysis, open problems	BoxLitE, FALCON, KB semantics	1 min

1

Hierarchy embeddings

Order and box embeddings
Hyperbolic space
RegD

Our first structured case: a hierarchy



A small taxonomy (example from RegD, KR 2026)

What must the geometry capture?

$C \sqsubseteq D$: "C is a subclass of D" (is-a)

- **Asymmetry**
  dog $\sqsubseteq$ animal, animal $\not\sqsubseteq$ dog
- **Transitivity**
  dog $\sqsubseteq$ animal $\sqsubseteq$ living thing
  $\Rightarrow$ dog $\sqsubseteq$ living thing

- **Where they appear:** plant taxonomy and product categories
  e.g. running shoes $\sqsubseteq$ shoes $\sqsubseteq$ footwear
- **Beyond trees:** a class may have multiple parents (a DAG)

Goal: embed a hierarchy while preserving its is-a relations

Order embeddings[4]: from points to nested regions

in depth

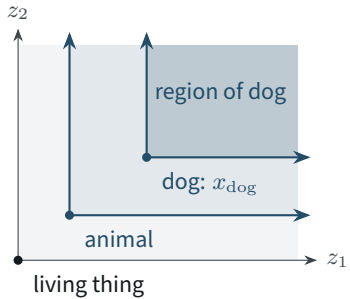
Definition: a region with apex x_u

Each concept u is embedded as an **upper orthant** R_u with apex $x_u \in \mathbb{R}^n_+$:

$$R_u = \{z \in \mathbb{R}^n_+ : z_i \geq (x_u)_i \text{ for all } i\}.$$

Encode $u \sqsubseteq v$ by inclusion:

$$R_u \subseteq R_v.$$



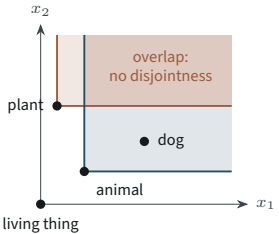
Transitivity comes from set inclusion:

$R_{\text{dog}} \subseteq R_{\text{animal}} \subseteq R_{\text{living thing}} \Rightarrow R_{\text{dog}} \subseteq R_{\text{living thing}}$.

Vendrov et al., ICLR 2016 [4], §2.1; region (cone) view: Vilnis et al., ACL 2018 [5], §3.2

Order embeddings[4]: two limitations

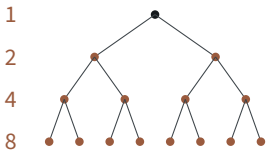
1 No disjointness



Animal and plant always overlap.

2 Crowding

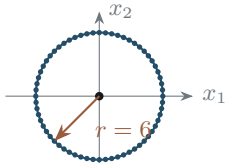
Binary tree (unit edges)



depth 3: $2^3 = 8$ nodes

depth 6: $2^6 = 64$ nodes

2D Euclidean space



64 nodes on the circle

$2\pi r \approx 37.7$

A b-ary tree: b^r **nodes** at depth r (unit edges).

In $\mathbb{R}^n$: sphere area $\propto r^{n-1}$.

Order embeddings: Vendrov et al., ICLR 2016 [4]; no disjointness: Vilnis et al., ACL 2018 [5], §4.2; crowding: Nickel & Kiela, NeurIPS 2017 [7], §2

τBox[16]: extending order embeddings with boxes

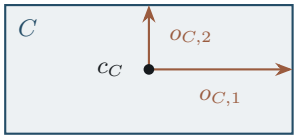
Box definition

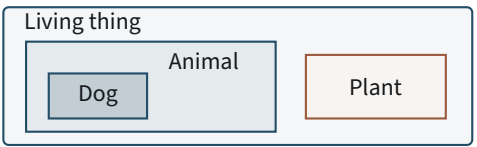
Each concept C is an axis-aligned box:

$$\mathrm{Box}(C) = [\, c_C - o_C,\ c_C + o_C \,].$$

$c_C \in \mathbb{R}^n$: **centre**

$o_C \in \mathbb{R}^n_{\geq 0}$: **offset** (half-width in each dimension)





What boxes fix

Containment: encode is-a.

Disjointness: encode empty intersection.

Remaining limitation

Euclidean space still suffers from **crowding**.

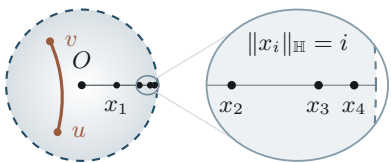
Box embeddings: Vilnis et al., ACL 2018 [5]; τBox: Boratko et al., NeurIPS 2021 [16]

Hyperbolic space: the Poincaré ball

Definition: the Poincaré ball $\mathbb{B}^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$

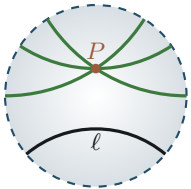
$$d_{\mathbb{H}}(u, v) = \operatorname{arcosh}\left(1 + \frac{2\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)}\right)$$

Shortest paths



Orange curve: shortest path from u to v
$\|x\|_{\mathbb{H}} := d_{\mathbb{H}}(O, x)$ (distance from O)
Boundary: **infinitely far**

Parallel lines

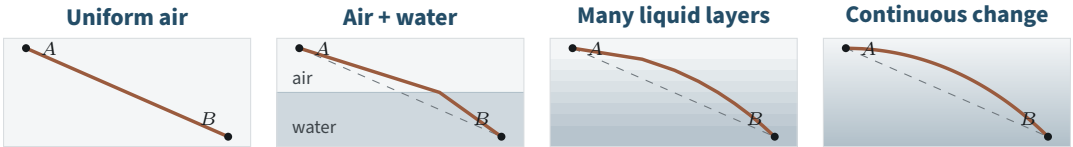


Through P: **infinitely many** lines never meet ℓ
(Euclidean space: exactly one)

Poincaré ball model; distance as in Nickel & Kiela, NeurIPS 2017 [7], §3, Eq. 1

Why can a “straight line” look curved?

An optical analogy for why a shortest route can look curved.



Orange: fastest light path

Dashed: straight shortcut

Light: fastest route

Changing speed can make the fastest route bend.

Hyperbolic space: the same intuition

In the Poincaré ball picture, “density” increases towards the boundary, so shortest paths can look curved.

> An intuitive motivation only: not an equivalent model or a definition of hyperbolic space.

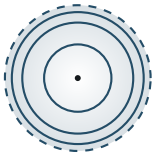
Optical analogy: Feynman Lectures I, Ch. 26; Poincaré model: Nickel & Kiela, 2017 [7]

Why hyperbolic space? Room that grows like a tree

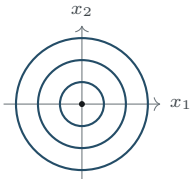
Crowding comes from $\mathbb{R}^n$: trees grow exponentially, Euclidean room only polynomially



Tree
2^ℓ nodes at level ℓ



Poincaré ball
$L(r) = 2\pi \sinh r$



Euclidean plane
$L(r) = 2\pi r$

× **Euclidean $\mathbb{R}^n$: polynomial**

Sphere of radius r: area $\propto r^{n-1}$.

Hyperbolic (curvature -1): exponential

Sphere area $\propto \sinh^{n-1} r \sim e^{(n-1)r}$.

Nickel & Kiela, NeurIPS 2017 [7], Section 2

Any tree fits in two hyperbolic dimensions

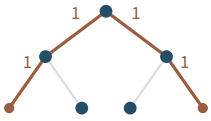
A tree as a **distance space**: vertices are points, path lengths are distances

Theorem (Sarkar, GD 2011 [8])

For every finite weighted tree T and every $\varepsilon > 0$, there are an embedding $f : V(T) \to \mathbb{H}^2$ and a global scale $\tau > 0$ such that

$$\tfrac{1}{1+\varepsilon}\, d_T(u,v) \;\le\; \tfrac{1}{\tau}\, d_{\mathbb{H}}\big(f(u), f(v)\big) \;\le\; d_T(u,v).$$

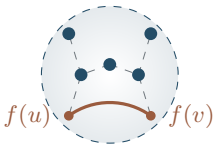
(a) Tree metric $(V(T), d_T)$



$d_T(u,v) = 1+1+1+1 = 4$

f

(b) Embedding in $\mathbb{H}^2$



$d_{\mathbb{H}}(f(u), f(v))$

Sarkar, GD 2011 [8]. Schematic mapping, not the theorem's exact construction.

Poincaré embeddings[7]: an asymmetric score

Score function ($\alpha > 0$)

$$\mathrm{score}(u \sqsubseteq v) = -\big(1 + \underbrace{\alpha(\|v\|_{\mathbb{H}} - \|u\|_{\mathbb{H}})}_{\text{asymmetry term}}\big) \quad \underbrace{d_{\mathbb{H}}(u, v)}_{\text{hyperbolic distance}}$$

$\|x\|_{\mathbb{H}} := d_{\mathbb{H}}(O, x)$: **hyperbolic norm** (distance from the origin).

- $d_{\mathbb{H}}(u, v) = d_{\mathbb{H}}(v, u)$: distance alone gives no direction.
- With the asymmetry term:

$$\|v\|_{\mathbb{H}} < \|u\|_{\mathbb{H}} \quad \Longrightarrow \quad \mathrm{score}(u \sqsubseteq v) > \mathrm{score}(v \sqsubseteq u).$$

The score favours **parents closer to the origin** than their children.

Hyperbolic-radius variant of Nickel & Kiela, NeurIPS 2017 [7], §4.3, Eq. 7

Poincaré embeddings[7]: WordNet results

Reconstruct the transitive closure of WordNet nouns: 743k is-a edges among 82k nouns

Method	Dimension	Space	MAP ↑	Mean rank ↓
Poincaré embeddings[7]	5	Hyperbolic (Poincaré ball)	**0.82**	**4.9**
Euclidean point baseline	200	Euclidean	0.17	1157

MAP ↑

How well related nouns are ranked above unrelated ones.
1 is best.

Mean rank ↓

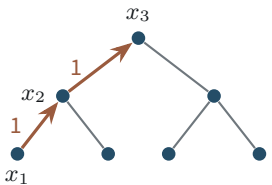
Average position of a related noun.
Lower is better.

> 40× fewer dimensions, far better reconstruction

Nickel & Kiela, NeurIPS 2017 [7], §4.1 and Table 1

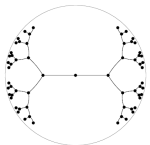
Hyperbolic embeddings keep distances, not order

Sarkar's theorem preserves the tree metric.
It does not define an is-a relation.



Arrows: child $\to$ parent





Poincaré disk

Preserved: distances

Sarkar: distortion at most $1 + \varepsilon$.

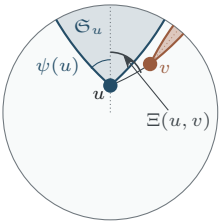
Not preserved: transitive

x_1 is a child of x_2, x_2 is a child of x_3.

$\not\Rightarrow$ x_1 is a child of x_3.

Sarkar, GD 2011 [8]. Visualisation: Stanford DAWN

Entailment cones[14]: hierarchy through nested regions



$\mathfrak{S}_u$ (blue), $\mathfrak{S}_v$ (orange) $\quad v \notin \mathfrak{S}_u$.

Cone geometry

Apex u Half-aperture $\psi(u)$ around the outward radial direction.

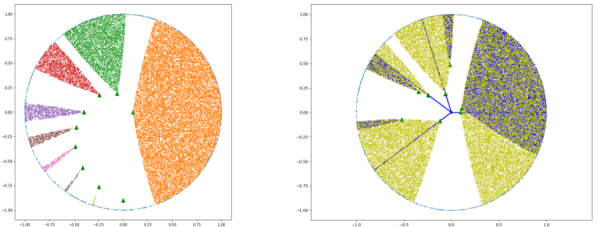
$$\psi(u) = \arcsin\left(K\frac{1-\|u\|^2}{\|u\|}\right), \quad K > 0$$

Entailment test

$\Xi(u, v)$: angle from the outward radius to the geodesic toward v.

$$E(u, v) = \max\big(0, \Xi(u, v) - \psi(u)\big)$$

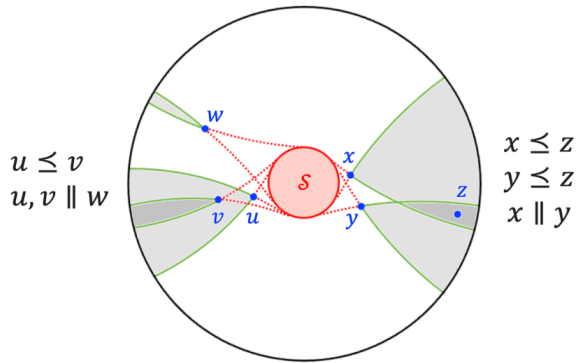
$E = 0$ inside $\quad E > 0$ outside $\quad$ Nesting gives transitivity.



Cone regions and an embedded hierarchy

Ganea et al., ICML 2018 [14], Eqs. 24–28, 33; Fig. 2. Left: $K = 0.2$, $\|u\| = 0.25$; $\|u\|$ is the Euclidean coordinate norm.

Shadow cones[15]: inspired by light and shadow



Original figure: $u \preceq v$ corresponds to $v \sqsubseteq u$.

- **Different construction:** cones inspired by light and shadow.
- **Same principle:** hierarchy through nested regions.

Yu et al., Shadow Cones: A Generalized Framework for Partial Order Embeddings, ICLR 2024 [15], Fig. 1.

Remaining limitations, and our question

× **Remaining limitations of hyperbolic methods**

- Specific cone constructions: incompatible with other region embeddings
- Hyperbolic computation, Riemannian optimisation and high numerical precision (e.g. FP64 / double precision)

Riemannian SGD: Bonnabel, IEEE TAC 2013 [10]; Bécigneul & Ganea, ICLR 2019 [11]. Precision: Sala et al., ICML 2018 [9]; Yu & De Sa, NeurIPS 2019 [12]; Mishne et al., ICML 2023 [13].

Can **Euclidean regions** get **hyperbolic expressivity**?

Remaining limitations, and our question

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Can **Euclidean regions** get **hyperbolic expressivity**?

RegD[17]: a Euclidean framework

1. **Plug-and-play**: works with existing region methods (balls, boxes)
2. **Euclidean realisation**: hyperbolic expressivity, standard computation

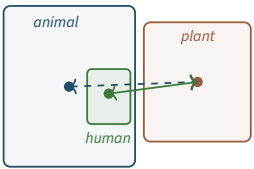
Goal: Euclidean regions + hyperbolic expressivity

What Euclidean regions lack: P1 and P2

Two key properties of hyperbolic space that Euclidean regions lack

P1: layer discrimination

A descendant can be arbitrarily farther from an outside concept than its ancestor



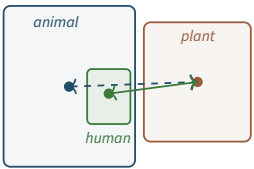
Euclidean centres: no gap guaranteed

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Two key properties of hyperbolic space that Euclidean regions lack

P1: layer discrimination

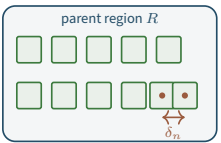
A descendant can be arbitrarily farther from an outside concept than its ancestor



Euclidean centres: no gap guaranteed

P2: child distinguishability

A parent can hold arbitrarily many children, pairwise far apart



Euclidean packing: children crowd together

Yang & Chen, KR 2026 [17], §3

RegD[17] (1): depth dissimilarity

in depth

Definition: depth dissimilarity for regions R_1, R_2

$$d_{\mathrm{dep}}(R_1, R_2) = g\left(\frac{\|P(R_1) - P(R_2)\|_p^p}{f(R_1)f(R_2)}\right)$$

P: parameters | $f(R) > 0$, region size $\to 0 \Rightarrow f(R) \to 0$ | g increasing, $g(0) = 0$.

E.g. $R = B(\mathbf{c}, r)$: $P(R) = (\mathbf{c}, r), \quad f(R) = r, \quad g(x) = x.$

Hyperbolic distance in the Poincaré ball $\mathbb{B}^n$

$$d_{\mathbb{H}}(u, v) = \operatorname{arcosh}\left(1 + \frac{2\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)}\right)$$

Yang & Chen, KR 2026 [17], §3.1, Prop. 1; Nickel & Kiela, NeurIPS 2017 [7], Eq. 1.

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Proposition 1: exact hyperbolic distance

For **Euclidean balls**, suitable f, g (with $p = 2$) make depth dissimilarity **exactly equal to hyperbolic distance** in one higher dimension.

Yang & Chen, KR 2026 [17], §3.1, Prop. 1; Nickel & Kiela, NeurIPS 2017 [7], Eq. 1.

RegD[17] (1): what depth dissimilarity guarantees

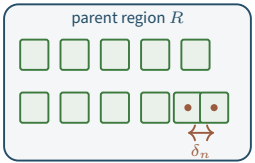
in depth

Theorem 1: d_{dep} realises P1 and P2 (balls and boxes alike)

P1: every small enough $R' \subseteq R_2$ has $d_{\mathrm{dep}}(R_1, R') > d_{\mathrm{dep}}(R_1, R_2) + \Delta$, for any $\Delta > 0$.

P2: every region contains arbitrarily many children, pairwise $d_{\mathrm{dep}} > M$, for any $M > 0$.

P2 example: fit more children, yet keep them distinguishable



Euclidean centres crowd together.

$$d_{\mathrm{dep}}(R_i, R_j) = g\left(\frac{\|P(R_i) - P(R_j)\|_p^p}{f(R_i) f(R_j)}\right)$$

Step 1. Shrink to fit. Make regions small enough to fit any number N inside the parent.

Step 2. Shrink again to separate. Even with tiny parameter gaps, shrink sizes further $\Rightarrow$ pairwise $d_{\mathrm{dep}} > \delta_n$.

Yang & Chen, KR 2026 [17], §3.1, Theorem 1. P2 example uses $g(x) = x$.

RegD[17] (2): boundary dissimilarity

Boundary dissimilarity: is R_2 inside R_1?

$$d_{\mathrm{bd}}(R_1, R_2) = \begin{cases} -\min_{\mathbf{x}_2 \in R_2} d(R_1^c, \mathbf{x}_2), & R_2 \subseteq R_1, \\ \max_{\mathbf{x}_2 \in R_2 \setminus R_1} d(R_1, \mathbf{x}_2), & \text{otherwise.} \end{cases}$$

Inside: minus the cost to move part of R_2 out of R_1.

Outside: the cost to move all of R_2 into R_1.

Yang & Chen, KR 2026 [17], §3.2, Def. of boundary dissimilarity and ball example.

RegD[17] (2): boundary dissimilarity

Boundary dissimilarity: is R_2 inside R_1?

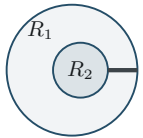
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Inside: minus the cost to move part of R_2 out of R_1.

Outside: the cost to move all of R_2 into R_1.

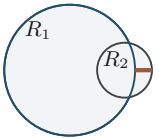
Example: balls $R_i = B(\mathbf{c}_i, r_i)$

$d_{\mathrm{bd}}(R_1, R_2) = \|\mathbf{c}_1 - \mathbf{c}_2\|_2 + r_2 - r_1$



$d_{\mathrm{bd}}(R_1, R_2) < 0$

contained



$d_{\mathrm{bd}}(R_1, R_2) > 0$

violation

$d_{\mathrm{bd}}(R_1, R_2) \leq 0 \iff R_2 \subseteq R_1$ (zero: contained and touching).

Yang & Chen, KR 2026 [17], §3.2, Def. of boundary dissimilarity and ball example.

RegD[17] (3): the score function

Poincaré embeddings[7]: points in $\mathbb{B}^n$

$$-\text{score}(u \sqsubseteq v) \;=\; \big(1 + \underbrace{\alpha\,(\|v\| - \|u\|)}_{\text{direction: norm}}\big) \cdot \underbrace{d_{\mathbb{H}}(u, v)}_{\text{hyperbolic distance}}$$

RegD[17]: regions in $\mathbb{R}^n$

$$E(u \sqsubseteq v) \;=\; \underbrace{d_{\text{bd}}(R_v, R_u)}_{\text{direction: inclusion}} \;+ \lambda \cdot \underbrace{d_{\text{dep}}(R_v, R_u)}_{\text{hyperbolic-like distance}}$$

Yang & Chen, KR 2026 [17], §3.3 and Table 1; Nickel & Kiela, NeurIPS 2017 [7], Eq. 7

RegD[17] (3): the score function

Poincaré embeddings[7]: points in $\mathbb{B}^n$

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RegD[17]: regions in $\mathbb{R}^n$

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- Lower is better.
- Predict $u \sqsubseteq v$ when $E \leq t$ (threshold tuned on validation).
- λ **tunes the distance term**: $\lambda = 0 \rightarrow$ pure inclusion
  $\lambda > 0 \rightarrow$ adds hyperbolic behaviour.

Yang & Chen, KR 2026 [17], §3.3 and Table 1; Nickel & Kiela, NeurIPS 2017 [7], Eq. 7

RegD[17] experiments (1): DAG inference

Train on basic edges, predict held-out inferred edges

F1 (%)	dims	best baseline	RegD[17]
Mammal	2 / 5	57.7 / 69.4	64.9 / 71.8
WordNet Noun	5 / 10	45.2 / 52.2	58.4 / 59.1
MCG	5 / 10	48.4 / 55.5	50.7 / 58.5
Hearst	5 / 10	41.6 / 46.8	42.8 / 49.6

- Baselines: entailment / shadow cones, order emb., τBox, ELBE, ELEm
- RegD best in **8/8** settings
- Largest gain $+13.2$ (Noun, $d{=}5$)

Yang & Chen, KR 2026 [17], Table 2

Hierarchy embeddings: a comparison

Method	Representation	Transitivity	Disjointness	No crowding (low dim.)	Euclidean computation
Order embeddings[4]	orthants in $\mathbb{R}^n_+$	✓	×	×	✓
τBox[16]	boxes in $\mathbb{R}^n$	✓	✓	×	✓
Poincaré embeddings[7]	points in $\mathbb{B}^n$	×	×	✓	×
Hyperbolic cones	cones in $\mathbb{H}^n$	✓	✓	✓	×
RegD[17]	any regions in $\mathbb{R}^n$	partial*	✓	✓	✓

* Full transitivity only for $\lambda = 0$ (pure inclusion, but then no hyperbolic room).

Summary of the methods above; RegD: Yang & Chen, KR 2026 [17], §3.3

2

EL-family ontology embeddings

EL-family normal forms
From points to regions
ELEm $\rightarrow$ boxes $\rightarrow$ TransBox

EL-family normal forms

From hierarchies to ontology axioms between concepts

Normal forms (templates)

NF1 $A \sqsubseteq B$

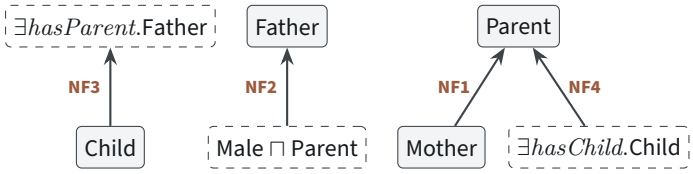
NF2 $A_1 \sqcap A_2 \sqsubseteq B$

NF3 $A \sqsubseteq \exists r.B$

NF4 $\exists r.B \sqsubseteq A$

A, A_i, B: concept names
r: role

Family example: a hierarchy of (complex) concepts



Arrows: $\sqsubseteq$ (sub $\rightarrow$ super)
Dashed boxes: complex concepts

Beyond basic EL: two role-axiom forms used here

Role inclusion: $r \sqsubseteq s$
Example: $hasMother \sqsubseteq hasParent$

Role-chain inclusion: $r_1 \circ r_2 \sqsubseteq s$
Example: $hasParent \circ hasParent \sqsubseteq hasGrandparent$

Normal forms: Baader et al., An Introduction to Description Logic, 2017 [1], Ch. 6; family ontology: Yang et al., WWW 2025 [23], Fig. 4a

Compare KG embeddings to ontology embeddings

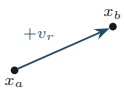
Basic idea: extend KG embeddings from **points** to **regions**

KG embedding (TransE[2])

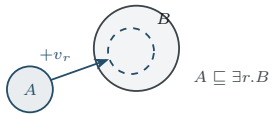
- Tuples $r(a, b)$ between **instances**
- Instance $\mapsto$ point

Ontology embedding

- Axioms $A \sqsubseteq B$ between **concepts**
- Concepts **include** each other $\Rightarrow$ concept $\mapsto$ **region**



KG: points



ontology: regions

TransE: Bordes et al., NeurIPS 2013 [2]

ELEm[18]: balls and translations

in depth

Concept $A \mapsto \mathrm{Ball}(A) := B(c_A, r_A)$ (centre, radius)
Role $r \mapsto$ translation vector v_r

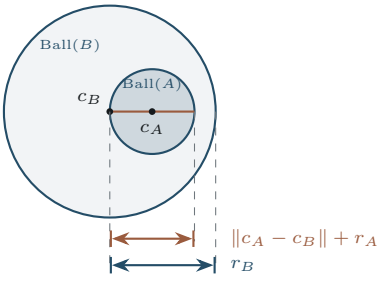
NF1 $A \sqsubseteq B$**:** $\mathrm{Ball}(A) \subseteq \mathrm{Ball}(B)$

$$\mathrm{Ball}(A) \subseteq \mathrm{Ball}(B) \iff \|c_A - c_B\| + r_A \leq r_B$$

Loss: $[\,\|c_A - c_B\| + r_A - r_B - \gamma\,]_+$ (margin γ)

Encode: $(\exists r.B)^{\mathcal{I}} = \mathrm{Ball}(B) - v_r$.

NF3 $A \sqsubseteq \exists r.B$: $\mathrm{Ball}(A) + v_r \subseteq \mathrm{Ball}(B)$.
NF4 $\exists r.B \sqsubseteq A$: $\mathrm{Ball}(B) - v_r \subseteq \mathrm{Ball}(A)$.



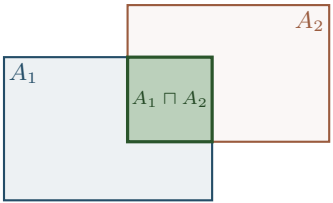
orange $\leq$ blue $\Rightarrow$ ball containment

NF2 $A_1 \sqcap A_2 \sqsubseteq B$: $\mathrm{Ball}(A_1) \cap \mathrm{Ball}(A_2)$ is generally not a ball $\Rightarrow$ **approximate** loss.
Zero loss need not imply $\mathrm{Ball}(A_1) \cap \mathrm{Ball}(A_2) \subseteq \mathrm{Ball}(B)$.

Kulmanov et al., IJCAI 2019 [18], §3.2, Eqs. 1–4

ELBE[20]: intersectional closure

brief



The box picture from ELBE

Roles use translations: $(\exists r.B)^{\mathcal{I}} = \mathrm{Box}(B) - v_r$.

NF2 $A_1 \sqcap A_2 \sqsubseteq B$

$$\mathrm{Box}(A_1) \cap \mathrm{Box}(A_2) \subseteq \mathrm{Box}(B)$$

Conjunction is represented exactly:

the intersection of two boxes is a box (or empty).

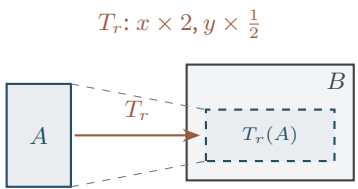
BoxEL[21]: roles as affine maps

Role definition (invertible case)

$$T_r(x) = D_r x + b_r, \quad D_r = \mathrm{diag}(d_{r,1}, \ldots, d_{r,n}), \quad d_{r,i} > 0; \qquad r^{\mathcal{I}} = \{(x, T_r(x)) : x \in \mathbb{R}^n\}.$$

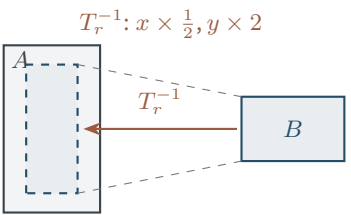
Encode $\exists r.B$ as $T_r^{-1}(\mathrm{Box}(B))$: take the preimage of B under the role map.

NF3 $A \sqsubseteq \exists r.B$



$T_r\big(\mathrm{Box}(A)\big) \subseteq \mathrm{Box}(B)$

NF4 $\exists r.B \sqsubseteq A$



$T_r^{-1}\big(\mathrm{Box}(B)\big) \subseteq \mathrm{Box}(A)$

Xiong et al., ISWC 2022 [21], §4.1, §4.4, Eqs. 9–10; adapted from Fig. 2(f), illustrative scales

Beyond functions: roles can be many-to-many

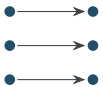
One transformation per role

ELEm[18] **/ ELBE**: one translation vector

$$x_b = x_a + v_r$$

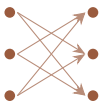
BoxEL[21]: one affine map per role.

$\Rightarrow$ Each input has **one output**.



One output per input

each input $\mapsto$ exactly one output



Many-to-many

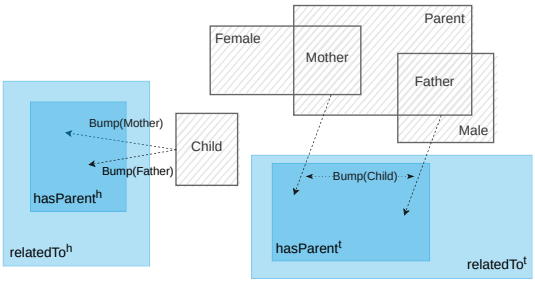
each input $\mapsto$ several outputs,
each output $\leftarrow$ several inputs
e.g. colleagueOf, hasParent

Next: **Box**^{2}**EL**[22] $\rightarrow$ head / tail boxes + bumps
TransBox[23] $\rightarrow$ boxes of translations

ELEm: Kulmanov et al., 2019 [18]; ELBE: Peng et al., 2022 [20]; BoxEL: Xiong et al., 2022 [21].

Box^{2}EL[22]: concept boxes and role position boxes

Each concept: a box and one learned bump.
Box^{2}EL: $A \mapsto (\mathrm{Box}(A), b_A)$



Stripes: concept boxes Blue: role head / tail boxes
Arrows: bumps

Each role: head and tail boxes.
$r \mapsto (H_r, T_r)$.

NF3 training: $A \sqsubseteq \exists r.B$

$$\mathrm{Box}(A) + b_B \subseteq H_r$$
$$\mathrm{Box}(B) + b_A \subseteq T_r$$

NF4 training: $\exists r.B \sqsubseteq A$

Step 1. Cover the existential region:
For any $x \in (\exists r.B)^{\mathcal{I}}$, NF3's head condition gives

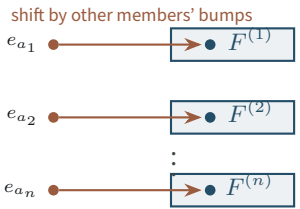
$$x + b_B \in H_r \;\Rightarrow\; x \in H_r - b_B.$$

Step 2. Require containment in A:
$H_r - b_B \subseteq \mathrm{Box}(A)$ (sufficient condition).

BoxE: the source of Box^{2}EL's point-and-bump idea

$F(a_1, \ldots, a_n)$ is a relation fact: F names the relation, and each a_i is a participating entity.

- **Example:** $F = \text{TeamMembers}$
  Each a_i is one team member.
- **Each member** a_i**:** a point e_{a_i} and one learned bump b_{a_i}.
- **Relation** F**:** one box $F^{(i)}$ for each argument position i.



One box per argument position

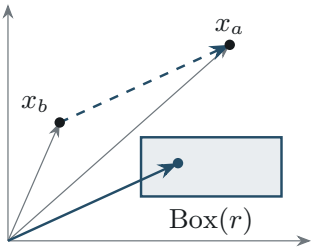
$F(a_1, \ldots, a_n)$: $e_{a_i} + \sum_{j \neq i} b_{a_j} \in F^{(i)}$ for every i. Other members' bumps only.

Abboud et al., NeurIPS 2020 (BoxE), §4, Eq. 1

TransBox[23]: roles as boxes of translations

in depth

A single translation vector → a **region of possible translations**



$r(a, b)$ iff $x_a - x_b \in \mathrm{Box}(r)$

A box for each role

Pair (a, b) satisfies r:
its displacement lies inside the role box

Many valid displacements

The same individual can relate to several others.

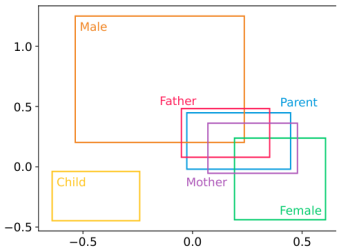
⇒ **many-to-many roles**

Concepts = regions of individuals
Roles = regions of translations

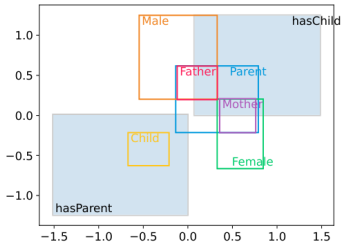
Yang et al., WWW 2025 [23], §4.1

TransBox[23]: an embedding example

Family ontology (TransBox paper): a child has a mother *and* a father, and Male and Female are disjoint



(b) ELBE



(d) TransBox

ELBE: Father and Mother overlap

TransBox: they are disjoint (roles are the **large** shaded boxes)

[screenshot] Yang et al., WWW 2025 [23], Fig. 4 (b, d)

TransBox[23]: sparse intersections

Box are sparse in high dimensions spaces.

Challenge: Theorem 4.5

Centers: $[-1, 1]^n$ Offsets: $(0, a_0]^n$ ($a_0 < 1$).

$$P(B \cap B' \neq \emptyset) \leq \left(a_0 - \tfrac{1}{4}a_0^2\right)^n \quad \Longrightarrow \quad n = 50,\ a_0 \uparrow 1 :\ P < 6 \times 10^{-7}.$$

Strategy: Extend the space.

Add an empty coordinate value:

$$\mathbb{R}^n \longrightarrow (\mathbb{R} \cup \{\emptyset\})^n$$

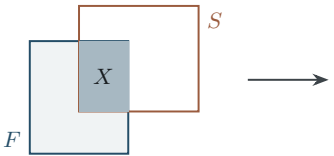
$$\widetilde{B} = \prod_{j=1}^{n} (I_j \cup \{\emptyset\})$$

- If one axis is disjoint, its coordinate becomes $\{\emptyset\}$. Other axes may overlap.
- An adapted loss for compute the inclusion in this case.

Analysis (1): composing a complex concept

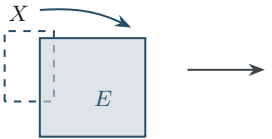
We can build **Person** $\sqcap \exists \mathit{hasChild}.(\textbf{Female} \sqcap \textbf{Student})$ in three steps

1 Intersect



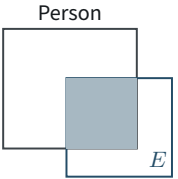
$X = \mathrm{Box}(F) \cap \mathrm{Box}(S)$
Female students

2 Apply the role



$E = X \oplus \mathrm{Box}(\mathit{hasChild})$
Those with a female student child

3 Intersect again



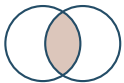
$\mathrm{Box}(\mathsf{Person}) \cap E$
The complete concept

Closure: every step returns a box $\Rightarrow$ any nesting can be built
Methods differ in steps 1, 2

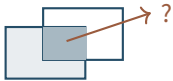
Yang et al., WWW 2025 [23], §3–4; $\oplus$: Minkowski sum; F = Female, S = Student

Analysis (2): conjunctions and NF4

Method	1 $C \sqcap D$	2 $\exists r.B$	How step 2 is built
ELEm[18]	✗	✓	translate a ball
ELBE[20]	✓	✓	translate a box
BoxEL[21]	✓	✓	affine preimage of a box
Box^{2}EL[22]	◐	✓	NF4: $H_r - b_B \subseteq \mathrm{Box}(A)$
TransBox[23]	✓	✓	box sum $\oplus$



ELEm, step 1: ball ∩ ball is a lens



Box^{2}EL, step 1: box intersection exists, but no bump for $C \sqcap D$

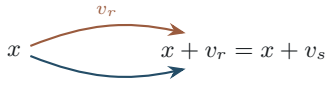
Kulmanov et al., 2019 [18]; Peng et al., 2022 [20]; Xiong et al., 2022 [21]; Jackermeier et al., 2024 [22]; Yang et al., WWW 2025 [23], §3

Analysis (3): how the methods handle role axioms

Compare the geometric condition with what each role representation can express.

Method	Role	$r \sqsubseteq s$	$r_1 \circ r_2 \sqsubseteq s$
ELEm, ELBE[18,20]	translation v_r	◐ only $v_r = v_s$	◐ only $v_{r_1} + v_{r_2} = v_s$
BoxEL[21]	affine map T_r	◐ only $T_r = T_s$	◐ only $T_{r_2} \circ T_{r_1} = T_s$
Box^{2}EL[22]	head, tail boxes	✓ $H_r \subseteq H_s, T_r \subseteq T_s$	◐ $H_{r_1} \subseteq H_s$, $T_{r_2} \subseteq T_s$
TransBox[23]	box of translations	✓ $\mathrm{Box}(r) \subseteq \mathrm{Box}(s)$	◐ $\mathrm{Box}(r_1) \oplus \mathrm{Box}(r_2) \subseteq \mathrm{Box}(s)$

ELEm / ELBE: one vector per role



$r \sqsubseteq s$ forces the two maps to coincide.

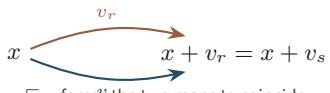
Jackermeier et al., WWW 2024 [22], §3.1–3.2; Yang et al., WWW 2025 [23], §4.1, Prop. 4.2; Mondal et al., AAAI-MAKE 2021 [19], §3.2.3 (EmEL++)

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Method	Role	$r \sqsubseteq s$	$r_1 \circ r_2 \sqsubseteq s$
ELEm, ELBE[18,20]	translation v_r	◐ only $v_r = v_s$	◐ only $v_{r_1} + v_{r_2} = v_s$
BoxEL[21]	affine map T_r	◐ only $T_r = T_s$	◐ only $T_{r_2} \circ T_{r_1} = T_s$
Box^{2}EL[22]	head, tail boxes	✓ $H_r \subseteq H_s, T_r \subseteq T_s$	◐ $H_{r_1} \subseteq H_s$, $T_{r_2} \subseteq T_s$
TransBox[23]	box of translations	✓ $\mathrm{Box}(r) \subseteq \mathrm{Box}(s)$	◐ $\mathrm{Box}(r_1) \oplus \mathrm{Box}(r_2) \subseteq \mathrm{Box}(s)$

ELEm / ELBE: one vector per role



$r \sqsubseteq s$ forces the two maps to coincide.

Can't handle transitive role ($r \circ r \sqsubseteq r$)

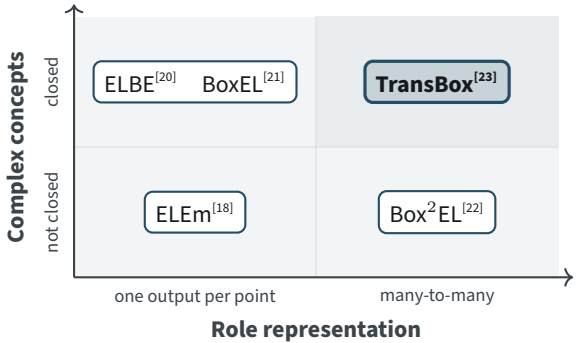
Box^{2}EL: $H_r \subseteq H_r, T_r \subseteq T_r$
Always true ⇒ every role is transitive.

TransBox: $\mathrm{Box}(r) \oplus \mathrm{Box}(r) \subseteq \mathrm{Box}(r)$
$\Rightarrow \mathrm{Box}(r) = \{\mathbf{0}\}$ or $\mathbb{R}^n$.

Jackermeier et al., WWW 2024 [22], §3.1–3.2; Yang et al., WWW 2025 [23], §4.1, Prop. 4.2; Mondal et al., AAAI-MAKE 2021 [19], §3.2.3 (EmEL++)

Analysis (4): capabilities at a glance

Two separate capabilities: composing complex concepts vs. many-to-many roles



One output per point

Translation or affine map:
each point $\mapsto$ **one** point

$\Rightarrow$ a child cannot reach both a mother and a father box

Many-to-many

Box^{2}EL: head / tail boxes
TransBox: a **box of translations**.

Right column: many-to-many roles and role axioms. **Training: see backup.**

ELEm [18], ELBE [20], BoxEL [21], Box^{2}EL [22], TransBox [23] (2019–2025)

3

Other works

Beyond the EL family
Theoretical analysis
Open problems

Other works: models and theory at a glance

brief

Different supported constructors — not a strict inclusion chain

DL-Lite$_{\mathcal{H}}$	EL family	$\mathcal{ALC}$ ($\neg$, $\sqcup$, $\forall$)
BoxLitE[32] KR'26 convex training (SOCP) ✓ weakly faithful	**DF-$\mathcal{EL}^{++}$**[33] KDD'26 fuzzy truth degrees	**FALCON**[24] preprint fuzzy models
	DELE[27] NAI'26 closure-aware negatives	**CatE**[25] NAI'25 concept lattice + order emb.
		Axis-aligned cones[26] JAIR'23 $\neg$ via polar cones

Theoretical analysis and results across ontology families

KB embedding semantics[29] KR'24
faithfulness, entailment closure, expressiveness

Strong faithfulness for ELH[30] TGDK'24
existence proof for convex geometric models

Discussion: hierarchy embeddings

Hyperbolic methods work well, but

- hard to implement: Riemannian geometry needs manifold-aware optimisation and numerical care.
- hard to reuse: special constructions (e.g. cones) bring method-specific order tests.

RegD[17]: a first step, hyperbolic in $\mathbb{R}^n$

- Transitivity is only partly supported: $\lambda = 0$ gives exact inclusion. $\lambda > 0$ adds depth but weakens that guarantee.
- Open score design: a sum $d_{\mathrm{bd}} + \lambda d_{\mathrm{dep}}$, a product (as in Poincaré), or another combination?

Hyperbolic optimisation: Mishne et al., ICML 2023 [13]; cones: Ganea et al., ICML 2018 [14]; RegD: Yang & Chen, KR 2026 [17], §3.3

Discussion: ontology embeddings

Problems

1. **Role axioms**: a role hierarchy is still weak.
   In particular, role chains $r_1 \circ r_2 \sqsubseteq s$.
2. **Beyond the EL family**: little work on $\mathcal{ALC}$, and mostly theoretical.
   Negation is the main problem
3. **Real-world applications**: move beyond axiom-ranking benchmarks.
   E.g., clinical ontology completion and ontology-guided biological prediction.
4. **Lack of transfer abilities**

Next (Chapter 3): language models bring in the labels

Ontology-embedding survey [3]; role-method comparison above; broader logics: FALCON [24], CatE [25].

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